

CSE306 project 2 report

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1 Overview

The objective of this project was to write a 2D fluid simulator using Voronoï cells and the Euler equations for incompressible liquids.

Originally, for N particles, we computed their Voronoï cells (regions in which they are the closest point by Euclidean distance). Afterwards, we had to balance the cells (have them have equal surface area) using a Power (Laguerre) diagram — each particle gets assigned a weight which basically transforms it into a circle, and the regions are made of points closest to a particle’s circle. Since the weights can be arbitrary, a particle’s region can be far from the particle itself based on the Euclidean distance. We had a functional to minimize, for which we used the LBFGS algorithm. For the 3rd part, we adapted the formula to include air, clipped every “water” cell with an “air” disk, and ran a particle simulation based on the rules of Physics: apart from gravity, we add a spring force which pulls particles towards the center of their Voronoï cell (which, again, can be far away Euclideanly).

The only optional work I did was the addition of the kNN-search for closest neighbours using KD trees through the nanoFlann library. Using this, instead of comparing against all the other particles, we can compare only to some “closest” neighbours (defined based on a distance not necessarily Euclidean). It was done outside of the lab.

2 Results

The following images (and mp4) were generated on my local machine running an Intel i7-13700H with 14 cores (6 fast with 2 threads each and 8 with only 1 thread, totalling 20 threads). The compilation flags relevant for performance were `-O3 -fopenmp -ffast-math`.

2.1 Lab 6

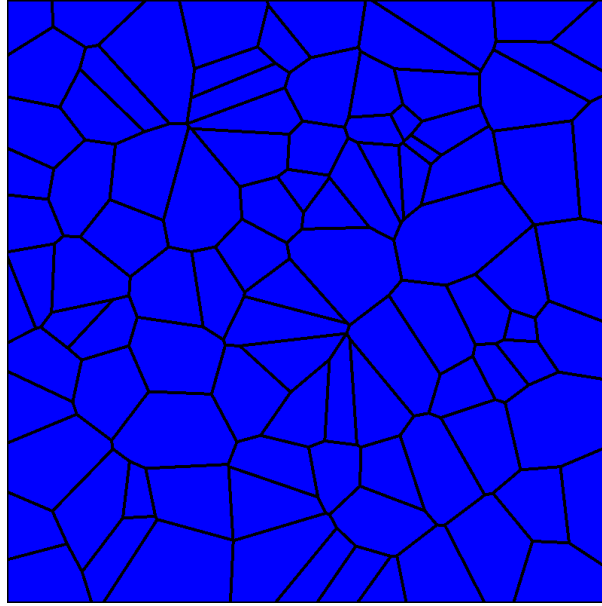


Figure 1: Result at the end of lab 1.

There is not much to say about this one. We select 100 points uniformly in the $[0, 1]^2$ square, and then for each, we clip a square (which is initially $[0, 1]^2$) by the bisector of the segment between that point and its closest 30 neighbours (less than this is needed for normal Euclidean distance, but once the weights are added it increases).

2.2 Lab 7

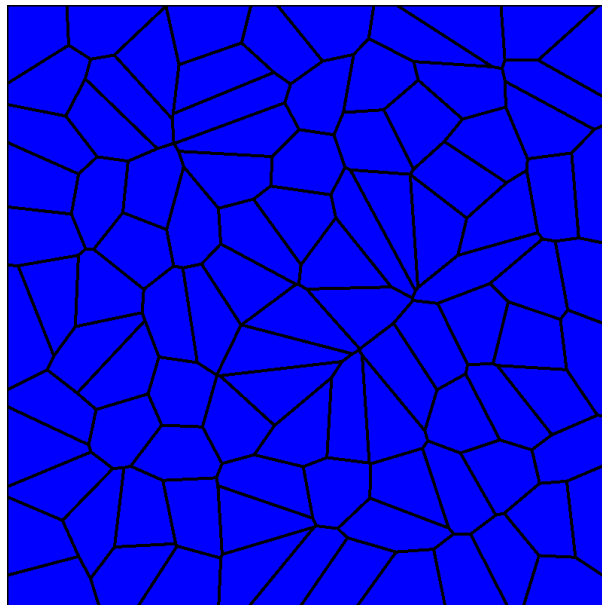


Figure 2: Result at the end of lab 2.

As we can see, the cells now have roughly the same surface. The only change was that the squared distance from a random point X to one of our points P_i is $\|X - P_i\|^2 - w_i$. We can “artificially” add this weight in the z coordinate of our points so that the kNN-search uses it. We then have to minimize the functional:

$$f(W) = - \left(\sum_i \int_{Pow_W(y_i)} (\|x - y_i\|^2 - w_i) dx + \sum_i \lambda_i w_i \right)$$

where $\lambda_i = \frac{1}{N_{particles}}$ is the desired “weight” (area in our case, since the whole image has area 1) of the cell. LBFGS just needs the current value of the function and its gradient, which are easily computable. At the end, using the obtained weights and our new distance logic, we generate the figure with almost equally large cells.

2.3 Lab 8

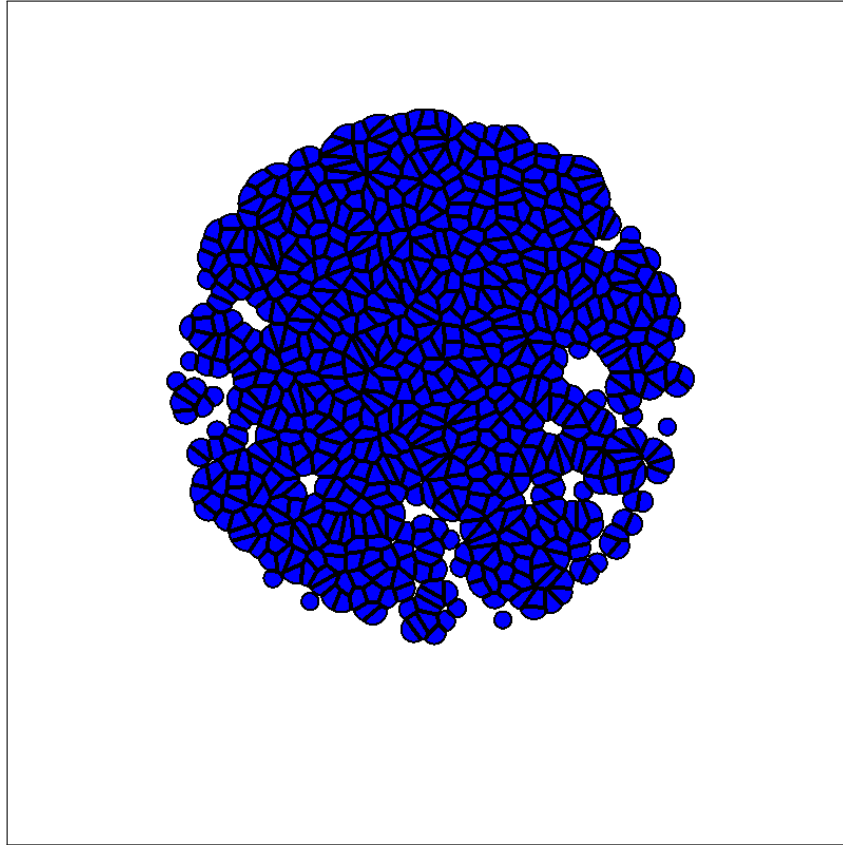


Figure 3: Starting configuration of liquid.

For this, we add a fictitious w_{air} at the end of the weights vector and adapt the formulas to consider the effect of air: first, the cells are clipped by a disk of size $\sqrt{w_i - w_{air}}$ if $w_i > w_{air}$, which effectively sets a lower bound on the cell size and stops it from growing until the edges of the image. We can imagine it as the implementation of air pressure. We then just simulate the particles moving for a small amount of time according to gravity and a force pulling them towards the center of their Power cell, then recompute

the Voronoï cells warmly (using the already-computed weights, since they are very good approximations for the weights at the next step so LBFGS will converge faster).

For some measurements, using 700 particles occupying a disk of radius 0.3 leading to $\pi 0.3^2 \approx 30\%$ water volume, 1000 frames and comparing with 200 neighbours, and clipping by a regular polygon with 50 vertices, the running time is:

```
real 12m22.022s
user 189m43.430s
sys 1m12.968s
```

This is ≈ 0.72 s per frame, which is under the desired outcome of 1s.

The video is available in the repository. It was generated with 1000 particles, 300 neighbours checked in the kNN-search, same radius 0.3, 60fps and $dt = 0.01$ so it moves pretty fast.

2.4 Potential improvements

As was suggested, this project could have been done in 3D or with a Newtonian solver (instead of a quasi-Newtonian one). Additionally, we could compute a dynamic number of neighbours instead of hard-coding a value (having 30 instead of 200 neighbours for 700 particles was a bug I had, which made me lose my mind as I had no idea why my animation was weird).

As for performance, it runs pretty smoothly, the only parameter that can reduce computation is the number of neighbours which could be fine-tuned. We can also reduce the number of particles, since the computation time is roughly proportional with it, but it would make the animation less of a liquid and more of a bubble tea.

2.5 Feedback on the course

The course overall was one of the most content filled CS courses I had in this program. However, because of the “work only in the TD” format, I often felt like I was implementing formulas I did not understand (the fact that TDs were sometimes right after the lectures did not help either, as I had no time to digest the content, but I think it is not in your control).

This makes debugging things much more difficult, because I often did not know which of the things I implemented was breaking. Also, especially for the first project, I feel like we could not have gotten something really nice out of it with only the amount of time spent during the TDs, considering the amount of optional things we could do to increase its quality. I know that the state of the world is pretty terrible right now considering all the AI development which makes users stop learning, and honestly I do not have any concrete solution to it.

Maybe having team-based project where contribution from each participant is quantified through the commits and the report, and a defense at the end? With teams of 3 there would be less than 20 teams, so less, better quality projects, that people have to explain.

The only downside is that the projects do not seem easily parallelisable, which would mean teammates would have to wait on other teammates. And so that people are not leaving anything towards the end of the semester, instead of doing everything during the TD, they could have to do it until the next TD, which I think is reasonable.

On another note, I really liked how the lecture notes were written, although I recommend putting them through an LLM to verify their correctness (for example, the gradient formula for the air weight was wrong in the lecture notes but right in the slides, the lecture notes use v_i to compute X_{i+1} in `time_step` in the lecture notes while the slides use v_{i+1}). I liked that the teaching team seemed very prepared to help the students and often knew what the problem was. The lectures were sometimes very full with material though, which made them harder to follow (especially since we did not use most of the formulas in the actual TD). A suggestion would be to keep them more abstract, since the lecture notes already contain the necessary information (I think).

I am open to talking more and could provide a channel of communication to my classmates (to make polls in our group chat, for example), if you would like me to. After we graduate though, I think people (myself included) are currently too busy. Thank you for being open to improvements :).